Design Thinking for innovation & Entrepreneurship (DTIE)

Section: 2

Project Batch: Project Batch-A

LunaBot: Autonomous Navigation for Lunar Habitats

An autonomous lunar rover system designed for ISRO's Smart India Hackathon 2025 built from the ground up around the real constraints of the Moon.

Concept render



DTIE | Department of Multi Disciplinary Innovation & Entrepreneurship

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Vision

A future where lunar habitats are supported by rovers that think for themselves, so astronauts and ground teams can focus on the mission instead of the machine.

Mission

Build a navigation stack designed around real lunar constraints: darkness and glare, loose regolith and communication delay.

Replace joystick-style control with high-level intent such as "patrol habitat".

Make every autonomous decision logged and auditable by ground operators.

Validate on a low-cost prototype before scaling to flight-grade sensors.

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The Challenge

Lunar rovers can't be driven in real time, and standard autonomy breaks down on the Moon's lighting and regolith.

Communication Delay

Robots must navigate lunar terrain without real-time human control a ~1.3 s one-way (~2.6 s round trip) delay makes remote correction impractical.

Extreme Lighting

Permanently shadowed regions create near-total darkness, while sunlit zones produce harsh glare both in the same mission area.

Loose Regolith

The lunar surface is covered in fine, powdery regolith that behaves very differently from solid ground causing slippage and unpredictable traction.



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Research & Industry Context

384,400	2.6 s	2023
Earth-Moon Distance (km)	**Round-trip Signal Delay**	**Chandrayaan-3 Landing**
Average distance to the Moon	Average communication delay	Recent milestone in lunar exploration

Facts

- Earth-Moon distance is about 384,400 km: ~1.3 s one-way, ~2.6 s round-trip signal delay, before processing time.
- Permanently shadowed regions near the lunar poles are of high interest because they may hold water ice, and they get no direct sunlight.
- Regolith is fine, abrasive and compressible, causing wheel slip and sinkage; slope alone does not predict traversability.
- Orbital imagery cannot show wheel-level hazards such as small rocks, ruts and soft patches.

Industry Context

- Renewed lunar activity: ISRO Chandrayaan-3 (2023, Pragyan rover), NASA Artemis programme, CNSA and private landers.
- Planned habitats and long-duration surface operations create demand for patrol, docking, logistics and EVA-support robots.
- Rover autonomy in shadowed, low-traction terrain is still a recognised open problem.

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Target User – Customer

Who are we solving this for?

Astronauts on EVA

Need a rover that does not require their attention during surface walks

Every second spent correcting a rover is a second not spent on mission tasks, in an environment with limited oxygen and power budgets

Ground Control Operators

~1.3 s one-way (~2.6 s round trip) communication delay makes real-time joystick control physically impossible

Operators need to issue high-level intent ("patrol habitat") rather than moment-to-moment commands

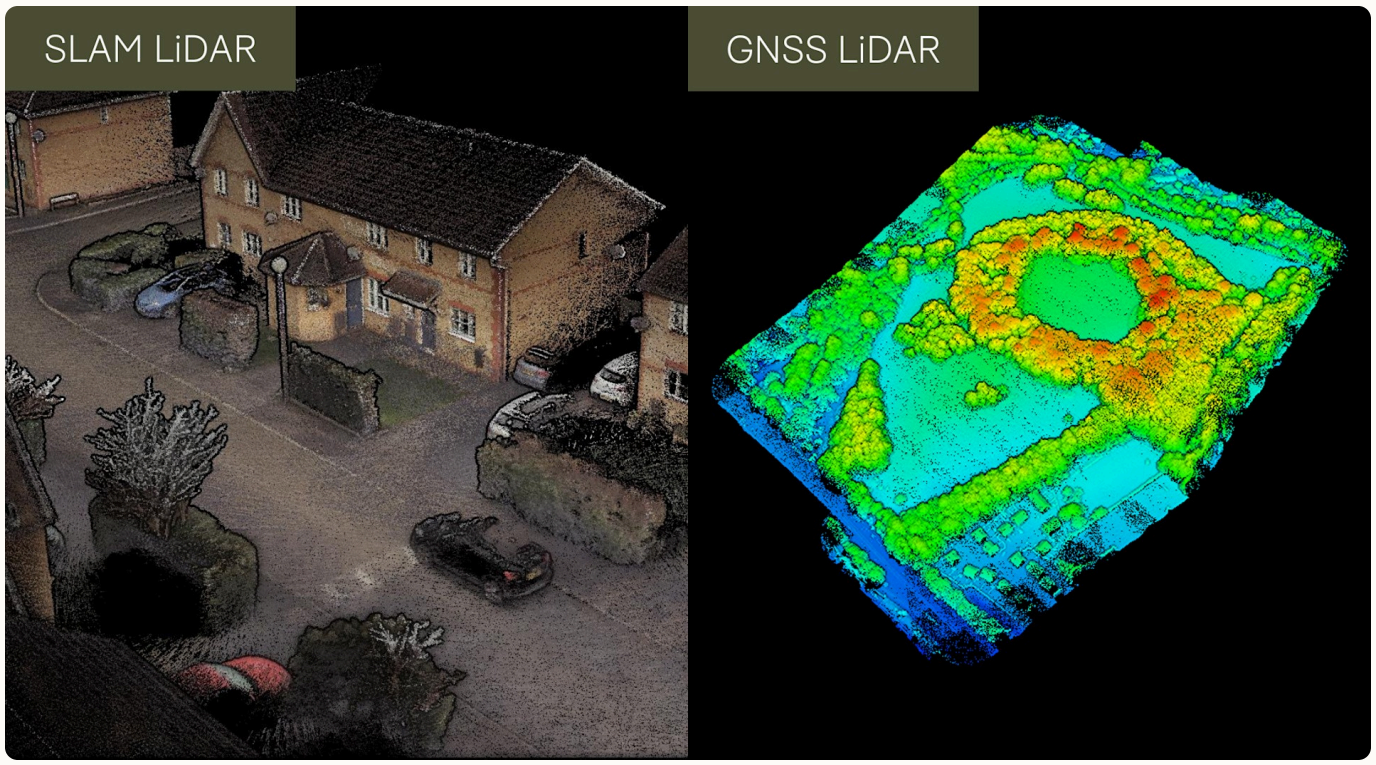
Habitat Safety Teams

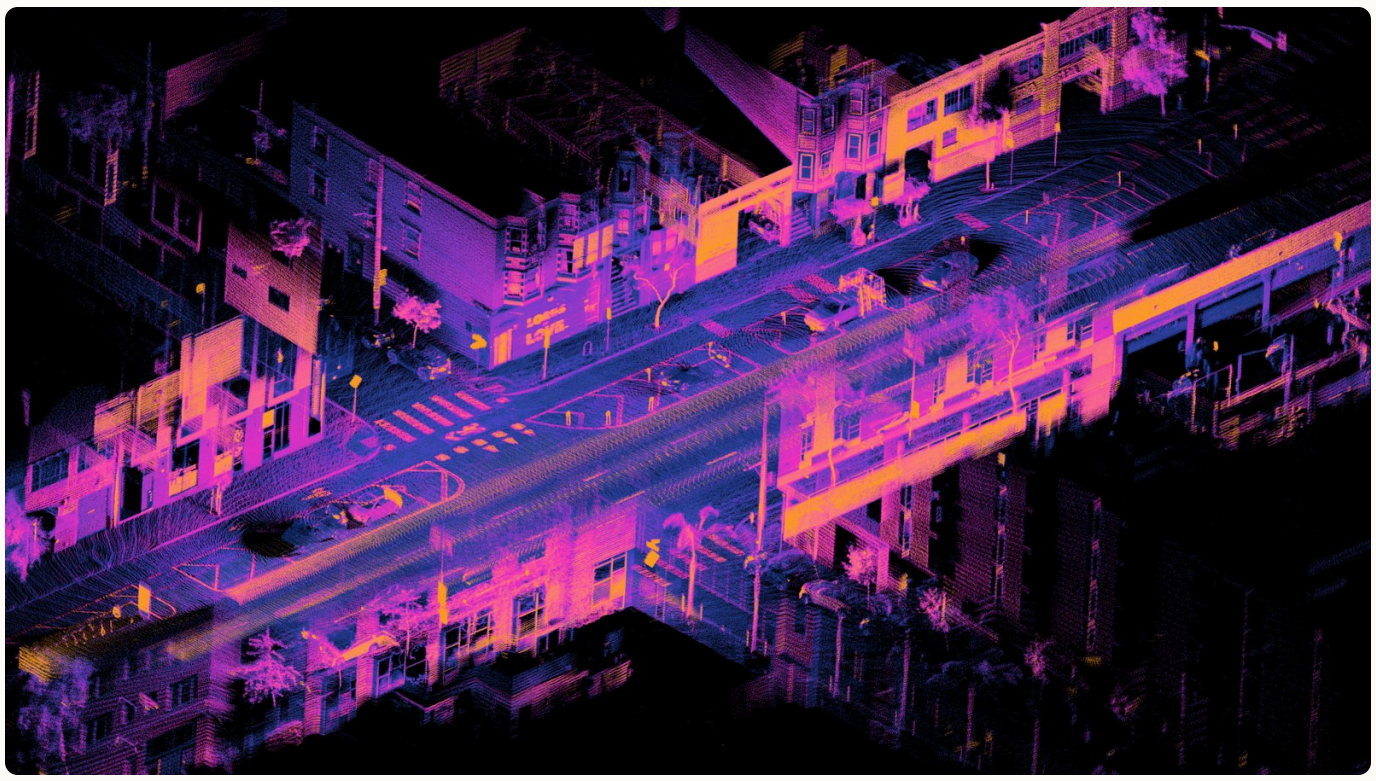
Need continuous perimeter awareness without dedicating crew time to manual patrols

Current rovers force a choice between slow, safe, human-supervised movement or fast, risky, unsupervised movement — neither is acceptable for real lunar habitat operations.

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Why Standard Solutions Fall Short





Vision-Only Systems Fail in Shadows

Camera-based navigation breaks down precisely where it's needed most in permanently shadowed craters and dark terrain.

Slope Checks Miss Regolith Dynamics

Simple geometry-based traversability ignores how wheels actually interact with loose, compressible lunar soil.

Human-in-the-Loop Dependency

Most systems still rely on operators for real-time correction an assumption that breaks under ~1.3 s one-way (~2.6 s round trip) communication delay.

Orbital Maps Are Too Coarse

Satellite imagery lacks the resolution to detect wheel-level hazards like small rocks, ruts, or soft patches.

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Idea Generation

Options Considered

Pure Vision/Camera-Based SLAM Navigation

Relies solely on stereo cameras and visual landmarks for localization and mapping

GNSS + Orbital-Map Path Following Only

Uses satellite positioning and pre-computed orbital maps for navigation without real-time sensing

Full Teleoperation with Predictive-Delay Buffering

Attempts to overcome communication delay through predictive algorithms and operator input buffering

Multi-Sensor Fusion (Camera + Thermal + LiDAR/Ultrasound) with Onboard Cost Mapping

Combines multiple sensor modalities with intelligent terrain scoring and autonomous decision-making (→ this became the LunaBot concept)

Swarm/Multi-Rover Relay Navigation

Uses multiple rovers to relay signals and distribute navigation tasks across a network

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Competitors / Benchmarks

Benchmark Comparison Table

Benchmark	What it is	LunaBot's difference
ISRO Pragyan (Chandrayaan-3)	Small rover with limited autonomy, mostly ground-directed	Habitat-scale missions with full onboard autonomy
CNSA Yutu-2	Long-lived rover with operator-planned drives	Less reliance on ground planning
Commercial lunar rovers (e.g. Astrolab, Lunar Outpost, ispace)	Payload carriers and utility rovers	Navigation stack built around shadows and regolith
NASA CADRE	Multi-rover cooperative autonomy demo	Single-rover, lower-cost approach

Comparison Grid

Approach	Works in shadow	Regolith-aware	Onboard autonomy	Low cost
Vision-only SLAM	No	No	Yes	Yes
GNSS + orbital maps	Yes	No	Partly	Yes
Teleoperation + buffering	Yes	No	No	Yes
Swarm relay	Partly	No	Yes	No
LunaBot	Yes	Yes	Yes	Yes

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Risks and Gap

No standard solution combines shadow-robust sensing, regolith-aware scoring and delay-tolerant autonomy in one stack.

Risk Mitigation Table

Risk	Impact	Mitigation
Sensor failure or noise in dust and extreme temperature	Wrong obstacle detection	Multi-sensor redundancy (camera + thermal + LiDAR/ultrasound)
Regolith behaves differently from the test arena	Prototype results may not transfer	Simulated regolith, slip-aware cost terms, higher-fidelity tests later
Onboard compute and power limits	Slower decisions, reduced range	Lightweight behaviour tree and cost map; emergency-return mode
Autonomy makes a wrong decision	Mission or hardware loss	Safe-stop behaviours and full decision logging for audit
Prototype uses ultrasonic stand-in for LiDAR	Limited range and resolution	Present as proof of concept; plan LiDAR upgrade
Funding and access to lunar-grade hardware	Slows path to flight	Phased roadmap, simulation first, partnerships

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Our Solution: The LunaBot Stack



Multi-Modal Perception

Stereo camera paired with thermal and depth sensors enabling reliable terrain sensing in both lit and shadowed environments.



Smart Terrain Scoring

Combines slope, surface roughness, and lighting conditions into a unified traversability score not just geometry.



True Autonomy

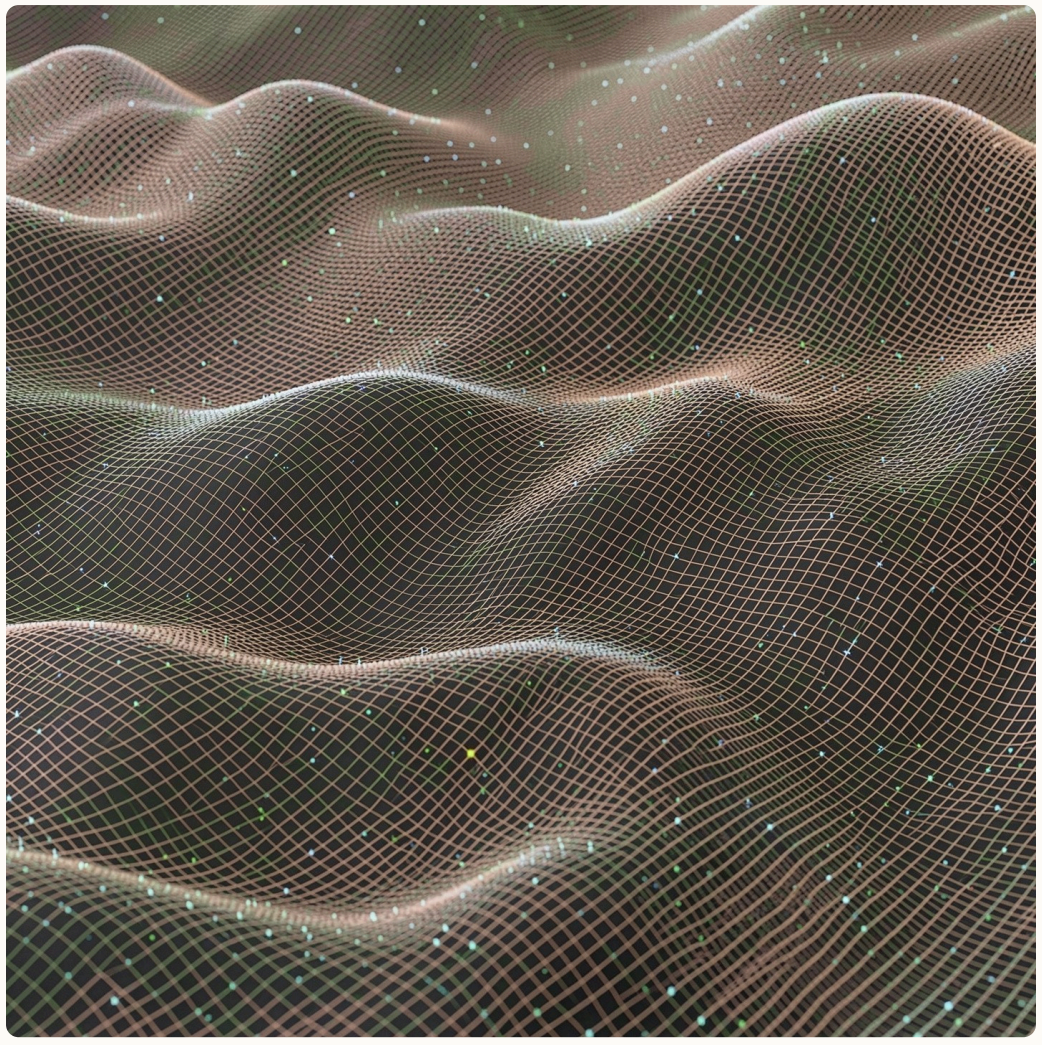
A behavior-tree state machine that operates fully independently under communication delay no ground operator needed for moment-to-moment decisions.

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Rough Prototype

What We Built Today

Current Prototype (Arduino-based)

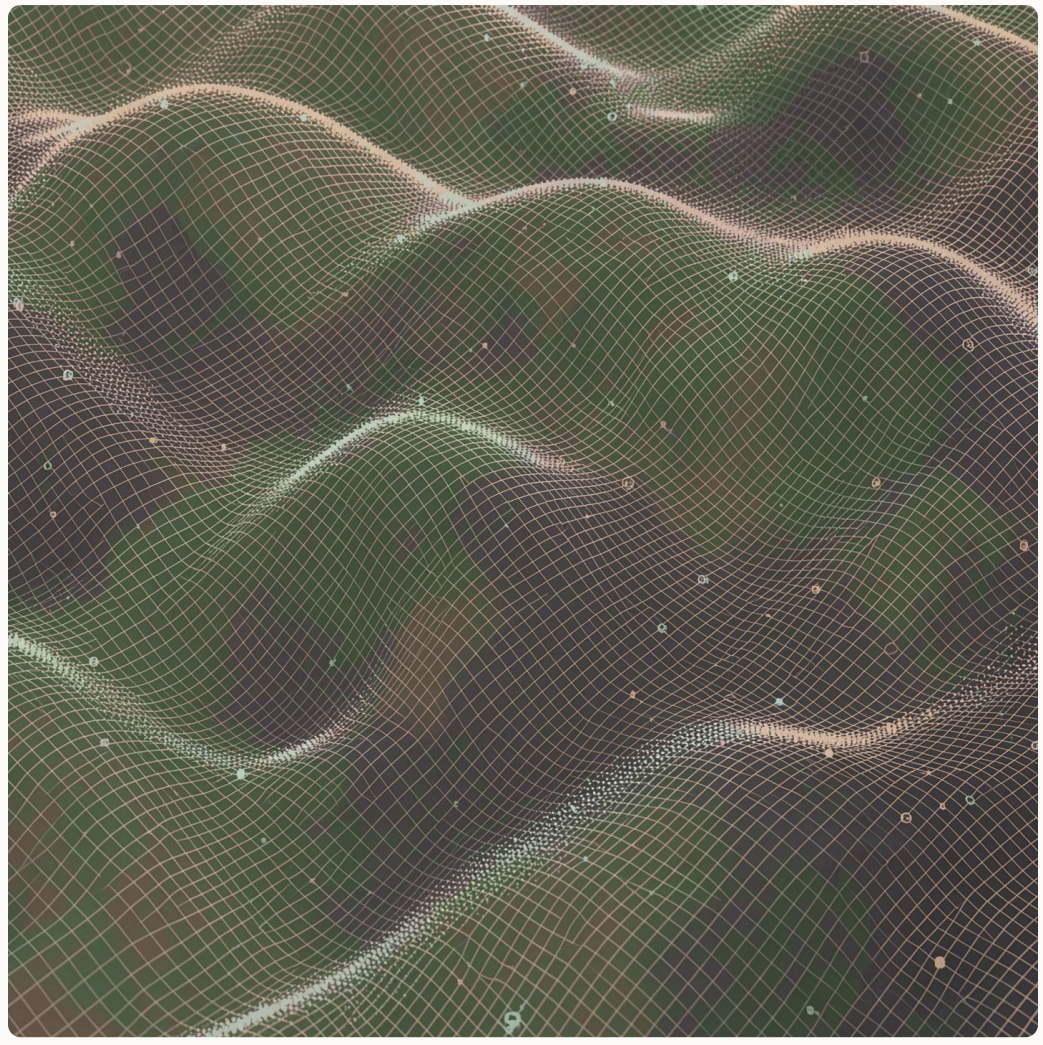


- Arduino Uno
- HC-SR04 ultrasonic sensor
- L298N motor driver
- 4 motors
- 2x18650 battery cells

Current build focuses on core motion and obstacle sensing with a simple, testable hardware stack.

Target Design

Target Design (Flight-grade sensors)

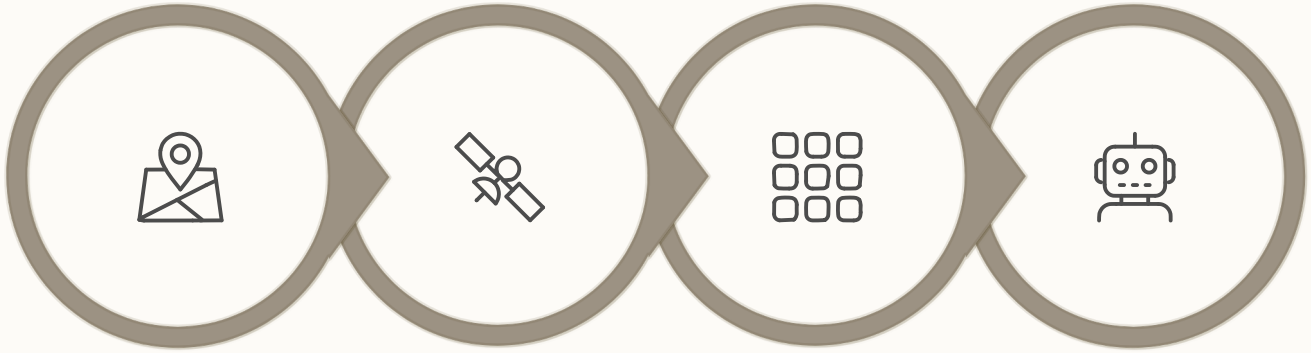


- Stereo camera
- Thermal sensor
- LiDAR
- Compute board (Jetson / Raspberry Pi class)
- Rugged chassis with encoders

The target platform expands sensing, compute, and durability for autonomous operation in harsher conditions.

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How It Works



Waypoints Perception Cost Map Decision

Closed-Loop Intelligence

Ground control sends high-level waypoints such as "patrol habitat" and LunaBot takes it from there.

The rover fuses orbital maps with live sensor data to continuously build and update its terrain understanding. A composite cost map steers it away from high-slip zones and shadowed hazards in real time.

Onboard decision-making handles obstacle avoidance and recovery without any ground input. Every decision is logged for ground operators to audit after the fact.

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Real-World Tasks

Autonomous Docking

Self-directed docking at charging stations without operator assistance.

Habitat Patrols

Scheduled perimeter patrols around the lunar habitat on a defined timetable.

EVA Support

Astronaut EVA support via "follow-me" mode acting as a mobile assistant during surface walks.

Emergency Return

Fastest-safe-return routing under power pressure — prioritizing survival over mission completion.

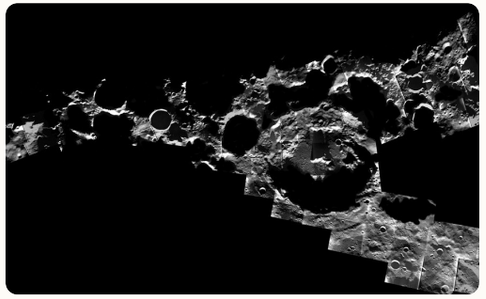
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A Day in the Life: LunaBot in Action



TASKING

Ground control sends "patrol habitat" and the rover starts with no further input.



DARKNESS

It enters a shadowed zone; thermal and depth sensing take over from the camera.

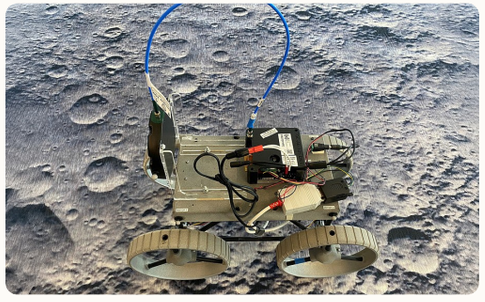
HAZARD

The cost map flags a soft patch and the rover reroutes on its own.



LOW POWER

Emergency Return picks the fastest safe route to the charging dock.



DOCKING

It docks autonomously and uploads its decision log for operators to audit.

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Lean Canvas Model

Problem	Solution	Unique Value Proposition
• Delay makes teleoperation impractical • Shadow and glare break vision-only navigation • Loose regolith defeats slope-only checks	• Multi-sensor fusion • Regolith-aware cost map • Behaviour-tree autonomy with auditable logs	• A rover built for the Moon as it actually is • Performing real habitat missions • Not point-to-point driving
Unfair Advantage	**Customer Segments**	**Channels**
• Constraint-first design (lighting, regolith, delay) • Integrated stack, not a single-sensor trick • Low-cost validated prototype	• Space agencies • Commercial lunar-habitat operators • Early adopters: research labs and analogue-mission teams	• Hackathons (SIH) and university research • Space-tech incubators and agency partnerships • Demos, papers, simulation releases
Key Metrics	**Cost Structure**	**Revenue Streams**
• Autonomous km without intervention • Obstacle-avoidance success rate • Mission task completion rate	• Sensors and compute • R&D and testing • Simulation, regolith test beds, certification	• Licensing the navigation software • Rover-as-a-service for habitat operations • Customisation and integration contracts

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Capital (Financial)

A. PROTOTYPE COST (What We Built)

Item	Approx. cost (INR)	Actual
Arduino Uno	600 - 1,000	600
HC-SR04 ultrasonic sensor	100 - 150	120
L298N motor driver	100-120	100
4-motor chassis kit with wheels	300-450	340
2x 18650 cells with holder	200-300	240
Wiring, switches, misc.	150-250	170
Total (approx.)	**about 1450-2500**	**1570**

B. UPGRADE TO FULL LUNABOT PROTOTYPE (Target Design)

Item	Approx. cost (INR)	Note
Compute board (Jetson / Raspberry Pi class)	6,000 - 40,000	Depends on the vision workload
Stereo / depth camera	8,000 - 25,000	Entry-level to mid-range
Thermal sensor / camera	5,000 - 30,000	Low-res array to a thermal camera
2D LiDAR	10,000 - 30,000	Entry-level 2D unit
Rugged chassis, motors, encoders	15,000 - 40,000	For loose-terrain testing
Simulated regolith arena and lighting	5,000 - 15,000	Sand/dust bed, dark and glare setups

Total estimate: ~60,000-200,000 INR

C. FUNDING PLAN

- Sources: college DTIE grant, hackathon prizes, incubator or agency grants, mentor or sponsor support.
- Total ask: INR 50,000
- Use of funds: Sensor upgrades, compute board, testing arena
- Revenue outlook: software licensing and rover-as-a-service (see Lean Canvas).

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Roadmap and Timeline

1. Concept

Done

Problem research, idea generation, concept selection (multi-sensor fusion + cost map + behaviour tree), SIH 2025 submission

2. Proof of concept

Done / in progress

Arduino obstacle-avoiding rover with ultrasonic sensing; controlled test zone

3. Perception upgrade

Next 3 - 6 months

Add stereo camera and thermal sensor; implement sensor fusion and SLAM

4. Cost map and autonomy

6 - 12 months

Composite traversability score, behaviour tree, task modes (patrol, docking, follow-me, emergency return)

5. Terrain validation

12 - 18 months

Test in simulated regolith and dark-lit arena; measure slip and success rate

6. Partnerships and pilot

18 - 36 months

Engage ISRO / incubators, simulation-based mission trials, seek funding

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Why This Matters

LunaBot reframes lunar robotics from a generic navigation problem to one built **directly around lunar constraints**: extreme lighting, regolith mechanics, and communication delay.

It delivers a rover that performs **actual habitat missions** not just point-to-point wandering.

> Designed for the Moon as it actually is not as a simplified simulation of it.



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